

Jenil Shingala

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EDUCATION

B.S. in Computer Science

Jan 2022 - Dec 2025

California State University, Sacramento • Major GPA 3.8 • Dean's Honor List

Coursework: Parallel & Concurrent Programming, Machine Learning, Artificial Intelligence, Operating Systems, Computer Architecture, Computer Networks, Data Structures & Algorithms

Certifications: NVIDIA-Certified Associate: AI Infrastructure & Operations (NCA-AIIO) (NCCL, DDP, Tensor Parallel, RoCE/InfiniBand, Multi-GPU) [LINK](#);

NVIDIA-Certified Associate: Generative AI LLMs (NCA-GENL) [LINK](#)

EXPERIENCE

AI Software Engineer Intern (Project Lead)

May 2025 - Dec 2025

Advanced Integration & Controls, Sacramento, CA

- Led design and deployment of a **production agentic RAG system**, grounding LLM responses in verified documents and improving accuracy by **96%** through **AI infrastructure** optimization.
- Built a **retrieval pipeline** using **Azure AI Search**, **vector embeddings**, and **Blob Storage** for high-performance semantic search and inference, raising daily engineer productivity by **87%**.
- Built **API-connected automations** (Bill of Materials, PCN, Monday.com tools) across **5+ data sources**, including an **MCP integration** to **Monday.com** and **Power Automate** flows.

ML/Software Engineer

Sep 2024 - May 2025

SentrySight LLC, Sacramento, CA

- Deployed a **full-stack AI app** with a **React** frontend and **AWS** backend (RDS, LightSail), serving real-time **computer-vision inference** at **<50ms latency**.
- Built a two-model **deep-learning pipeline** (YOLOv11n + transfer-learned CNN), achieving **91% CNN accuracy** and **94% face-recognition accuracy** across a **500+ person database**.
- Owned the full lifecycle from design through **CI/CD** deployment, building a **QA framework** with **15+ automated checks** and human-in-the-loop review to prevent quality drift before release.

Embedded Engineer Intern

Aug 2021 - Dec 2021

Goldfield Weighting Solution

- Wrote **C++ embedded firmware** for industrial systems, applying **object-oriented design patterns** at the **hardware-software** boundary.
- Worked with a cross-functional team of **4** on **hardware integration**, connecting firmware to physical scale **hardware**.
- **Tested** and **debugged** firmware end-to-end, isolating defects between **hardware and software** layers.

PROJECTS

AI Hackathon 2024 @ Hornet Hacks: Weapon and Face Detection | YOLOv11n · OpenCV · CNN · Inception v3 · Transfer Learning · CUDA. [GITHUB](#)

- Built a two-model **computer-vision** pipeline: **YOLOv11n** for object detection and a custom **CNN** via **transfer learning** from **Inception v3** for face recognition, achieving **94% detection accuracy**.
- Reduced false positives by **87%** with dual-model agreement, processing at **<50ms latency** across a **500+ individual database** with **91% CNN accuracy**.

Network Intrusion Detection System | Python · FCNN · CNN · Keras Tuner · TensorFlow. [GITHUB](#)

- Built a **deep-learning** intrusion-detection model (FCNN/CNN in **TensorFlow**) reaching **97% accuracy** on imbalanced network-traffic data.
- Ran **Keras Tuner** hyperparameter search and evaluated with **precision, recall, F1**, and confusion-matrix analysis to handle class imbalance.

AI Hackathon 2025 @ Hornet Hacks: Ticker Skimmer | Python · Bidirectional LSTM · RNN · NLP · Gradio. [GITHUB](#)

- Built a stock predictor combining Reddit sentiment (**NLP**) with a **bidirectional LSTM** time-series model and a **JIT optimization engine**, served through a real-time **Gradio** UI.
- Used **5-day time-step** sequences with **Z-score normalization** and **EarlyStopping** to capture pattern reversals from both past and future context.

FlashAttention GPU Kernel (from scratch) | Triton · PyTorch · CUDA · GPU Kernels · Online-Softmax. [GITHUB](#)

- Implemented **FlashAttention** as a fused **Triton GPU kernel** using **online-softmax** streaming recurrence, cutting peak attention memory from **O(N²)** to **O(N)**, about **250x lower** at sequence length 8192.
- **Benchmarked** against PyTorch `scaled\_dot\_product\_attention` using **CUDA-event timing** and peak-memory profiling, confirming **~17MB vs ~4.3GB** peak memory at sequence length 8192; extending with **Triton autotuning** and causal masking.

TECHNICAL SKILLS

Deep Learning & AI: LLMs, GenAI, Agentic RAG, Deep Learning, CNN, YOLO, Computer Vision, Bidirectional LSTM/RNN, NLP, Transfer Learning, AI Infrastructure, Keras Tuner, Model Evaluation (Precision/Recall/F1)

Languages & Cloud: Python, C, C++, JavaScript, React.js, Flask, AWS (RDS, LightSail), Azure, Git, CI/CD, Agile/SDLC, System Architecture (OOP), Linux

ML Systems & GPU: PyTorch, TensorFlow, CUDA, CUDA C++, Triton, GPU Kernel Development, GPU Architecture, NCCL, Distributed Data Parallel (DDP), Tensor Parallel, FlashAttention, HPC

Distributed & Performance: Multi-GPU, RoCE/InfiniBand, Parallel Computing, Performance Optimization, Benchmarking, Memory Optimization, Triton Autotuning, Latency Optimization, Scalability